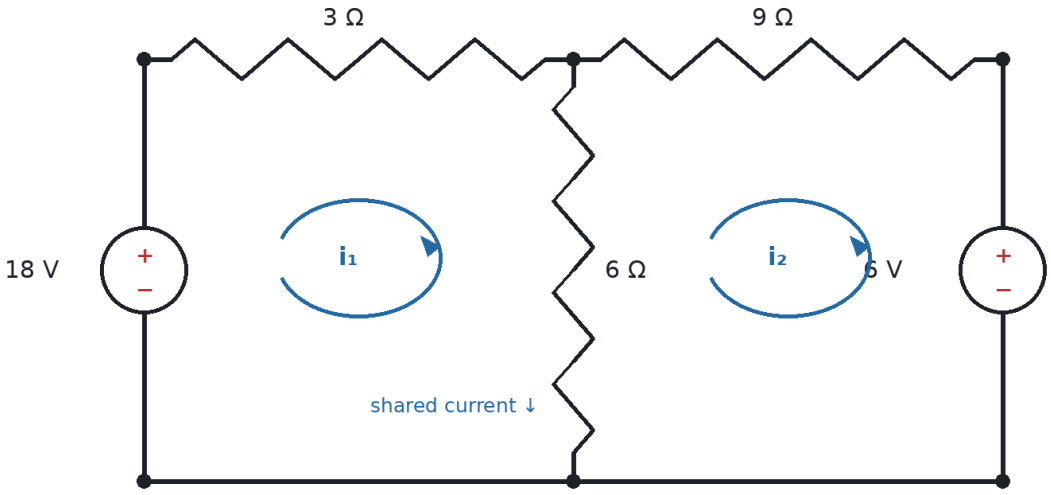
Mesh Questions

Question 1

Basic two-mesh currents. Find both mesh currents and the magnitude and direction of current in the shared resistor. The left top resistor is $3\,\Omega$, the shared resistor is $6\,\Omega$, and the right top resistor is $9\,\Omega$. The left and right sources are $18\,\text{V}$ and $6\,\text{V}$, respectively, both positive at the top.

Q03 — Two-Mesh Analysis



Two adjacent clockwise meshes share a $6\,\Omega$ vertical resistor. The left path has $3\,\Omega$ and a top-positive $18\,V$ source; the right path has $9\,\Omega$ and a top-positive $6\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $3i_1 + 6(i_1 - i_2) - 18 = 0$, $9i_2 + 6(i_2 - i_1) + 6 = 0$.

Thus, $(9)i_1 - 6i_2 = 18, \qquad -6i_1 + (15)i_2 = -6.$

Solving, $i_1 = \frac{26}{11} = 2.364\,\text{A}, \qquad i_2 = \frac{6}{11} = 0.5455\,\text{A}.$

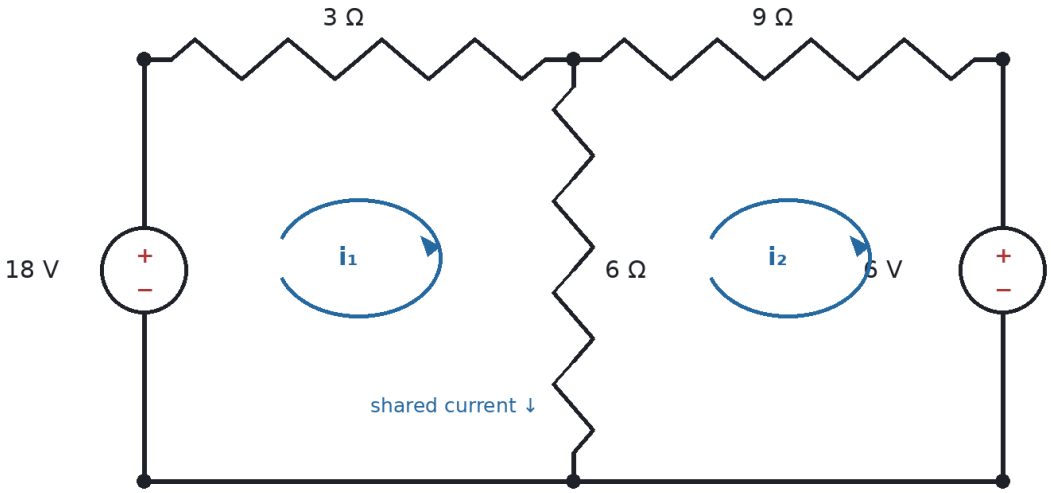
The shared current is $i_s = \frac{20}{11} = 1.818\,\text{A}$ downward .

A negative mesh current, if present, flows opposite its clockwise arrow.

Question 2

Basic source-power check. Find both mesh currents. Determine whether each voltage source absorbs or delivers power, and check total power. The left top resistor is $3\,\Omega$, the shared resistor is $6\,\Omega$, and the right top resistor is $9\,\Omega$. The left and right sources are $18\,\text{V}$ and $6\,\text{V}$, respectively, both positive at the top.

Q03 — Two-Mesh Analysis



Two adjacent clockwise meshes share a $6\,\Omega$ vertical resistor. The left path has $3\,\Omega$ and a top-positive $18\,V$ source; the right path has $9\,\Omega$ and a top-positive $6\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $3i_1 + 6(i_1 - i_2) - 18 = 0$, $9i_2 + 6(i_2 - i_1) + 6 = 0$.

Thus, $(9)i_1 - 6i_2 = 18, \qquad -6i_1 + (15)i_2 = -6.$

Solving, $i_1 = \frac{26}{11} = 2.364\,\text{A}, \qquad i_2 = \frac{6}{11} = 0.5455\,\text{A}.$

Using passive sign convention, $p_L = -(18)i_1 = 42.55\,\text{W}$ delivered , $p_R = (6)i_2 = 3.273\,\text{W}$ absorbed .

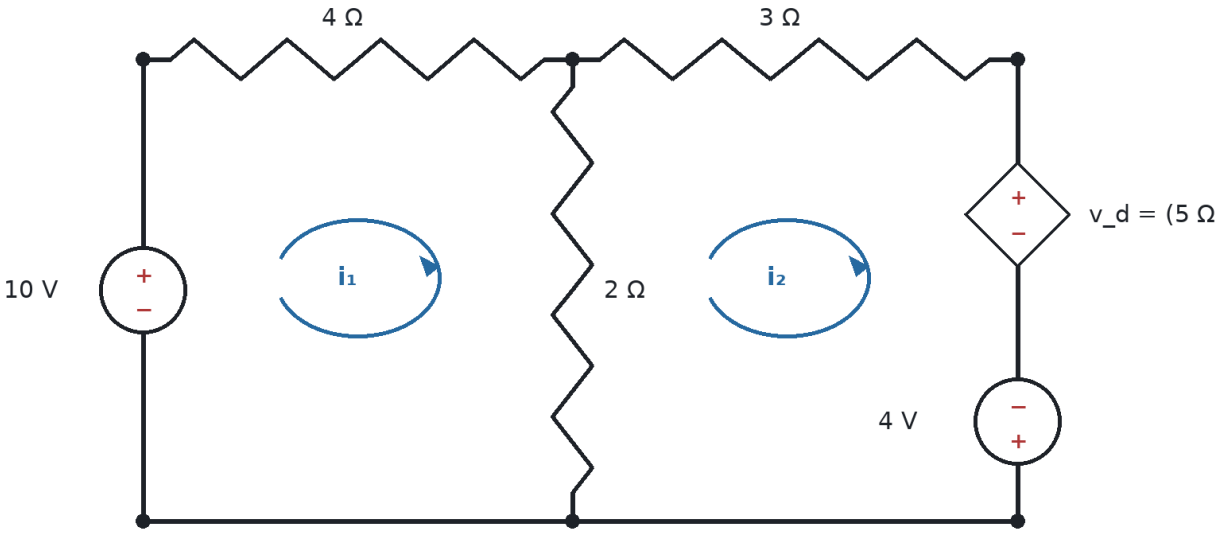
The resistors absorb $P_{\text{res}} = 3i_1^2 + 6(i_1 - i_2)^2 + 9i_2^2 = 39.27\,\text{W}$.

Therefore, total absorbed power is zero: $p_L + p_R + P_{\text{res}} = 0$.

Question 3

Dependent-source mesh currents. Find both mesh currents, the dependent-source voltage, and the shared current. Two meshes share a $2\,\Omega$ resistor. The left path has $4\,\Omega$ and a top-positive $10\,\text{V}$ source. The right path has $3\,\Omega$, a top-positive CCVS $v_d = (5\,\Omega)i_1$, and a $4\,\text{V}$ source positive at the bottom.

Q04 — Mesh Analysis with a Dependent Source



Clockwise meshes share $2\,\Omega$. The left path contains $4\,\Omega$ and a top-positive $10\,V$ source. The right path contains $3\,\Omega$, the top-positive CCVS $v_d = (5\,\Omega)i_1$, and a bottom-positive $4\,V$ source.

Solution

Define clockwise i_1 and i_2. Sum signed voltage drops. $-10+4i_1+2(i_1-i_2)=0, 2(i_2-i_1)+3i_2+5i_1-4=0$.

Therefore, $6i_1 - 2i_2 = 10, \qquad 3i_1 + 5i_2 = 4.$

Solving, $i_1 = \frac{29}{18} = 1.611\,\text{A}, \qquad i_2 = \frac{-1}{6} = -0.1667\,\text{A}.$

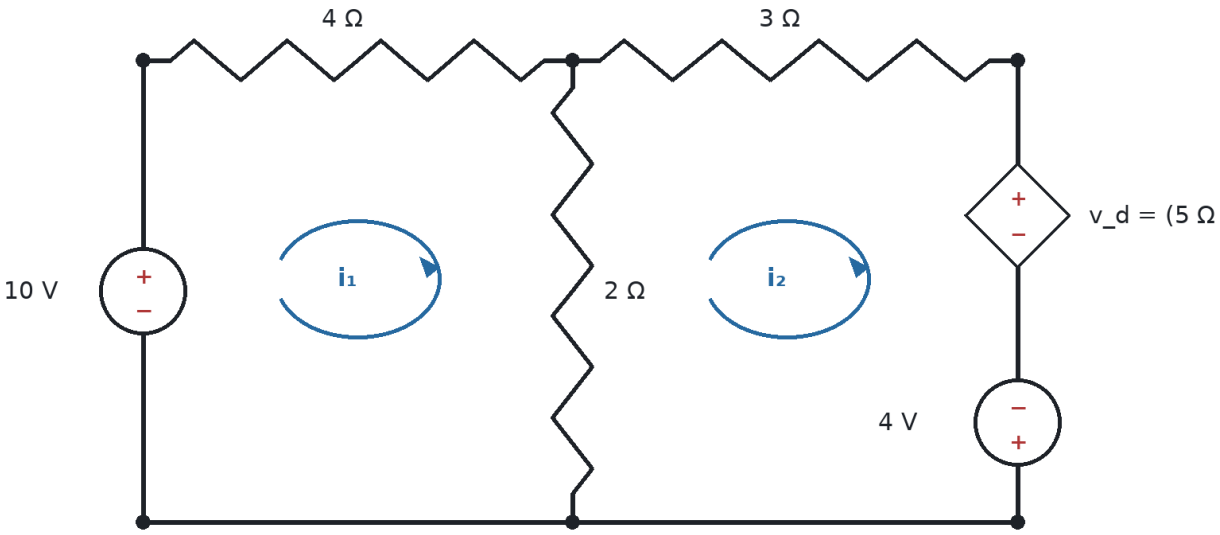
The actual right-mesh current is $0.1667\,\text{A}$ counterclockwise.

The dependent voltage and shared current are $v_d = 5i_1 = \frac{145}{18} = 8.056\,\text{V}$, $i_s = i_1 - i_2 = \frac{16}{9} = 1.778\,\text{A}$ downward.

Question 4

Dependent-source power. Find both mesh currents and determine the dependent source power using passive sign convention. Two meshes share a $2\,\Omega$ resistor. The left path has $4\,\Omega$ and a top-positive $10\,\text{V}$ source. The right path has $3\,\Omega$, a top-positive CCVS $v_d = (5\,\Omega)i_1$, and a $4\,\text{V}$ source positive at the bottom.

Q04 — Mesh Analysis with a Dependent Source



Clockwise meshes share $2\,\Omega$. The left path contains $4\,\Omega$ and a top-positive $10\,V$ source. The right path contains $3\,\Omega$, the top-positive CCVS $v_d = (5\,\Omega)i_1$, and a bottom-positive $4\,V$ source.

Solution

Define clockwise i_1 and i_2. Sum signed voltage drops. $-10+4i_1+2(i_1-i_2)=0, 2(i_2-i_1)+3i_2+5i_1-4=0$.

Therefore, $6i_1 - 2i_2 = 10, \qquad 3i_1 + 5i_2 = 4.$

Solving, $i_1 = \frac{29}{18} = 1.611\,\text{A}, \qquad i_2 = \frac{-1}{6} = -0.1667\,\text{A}.$

The actual right-mesh current is $0.1667\,\text{A}$ counterclockwise.

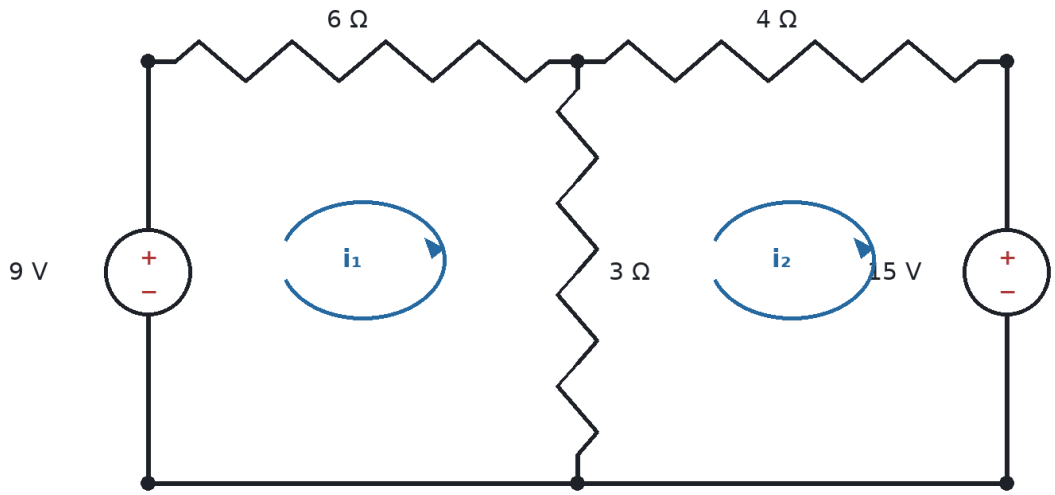
The CCVS current is i_2 downward into its positive terminal. Thus, $p_d = v_d i_2 = (\frac{145}{18})(\frac{-1}{6}) = -1.343\,\text{W}$.

The negative sign means the dependent source delivers $1.343\,\text{W}$.

Question 5

Opposing sources and direction. Find both mesh currents and the magnitude and direction of current in the shared resistor. The left top resistor is $6\,\Omega$, the shared resistor is $3\,\Omega$, and the right top resistor is $4\,\Omega$. The left and right sources are $9\,$V and $15\,$V, respectively, both positive at the top.

Opposing Sources



Two adjacent clockwise meshes share a $3\,\Omega$ vertical resistor. The left path has $6\,\Omega$ and a top-positive $9\,$V source; the right path has $4\,\Omega$ and a top-positive $15\,$V source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $6i_1 + 3(i_1 - i_2) - 9 = 0$, $4i_2 + 3(i_2 - i_1) + 15 = 0$.

Thus, $(9)i_1 - 3i_2 = 9, \qquad -3i_1 + (7)i_2 = -15.$

Solving, $i_1 = \frac{1}{3} = 0.3333\,$A, $\qquad i_2 = -2 = -2\,$A.

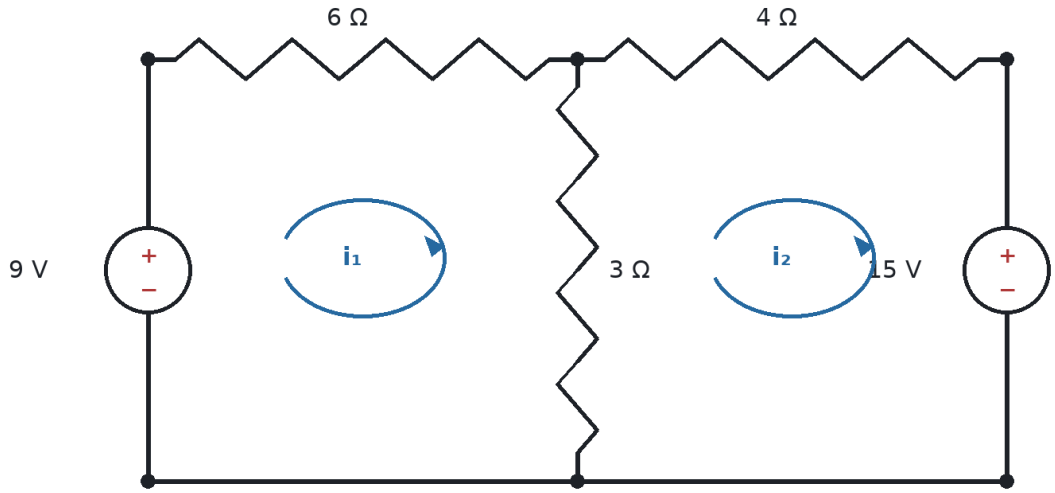
The shared current is $i_s = \frac{7}{3} = 2.333\,$A downward .

A negative mesh current, if present, flows opposite its clockwise arrow.

Question 6

Opposing-source power balance. Find both mesh currents. Determine whether each voltage source absorbs or delivers power, and check total power. The left top resistor is $6\,\Omega$, the shared resistor is $3\,\Omega$, and the right top resistor is $4\,\Omega$. The left and right sources are $9\,\text{V}$ and $15\,\text{V}$, respectively, both positive at the top.

Opposing Sources



Two adjacent clockwise meshes share a $3\,\Omega$ vertical resistor. The left path has $6\,\Omega$ and a top-positive $9\,V$ source; the right path has $4\,\Omega$ and a top-positive $15\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $6i_1 + 3(i_1 - i_2) - 9 = 0$, $4i_2 + 3(i_2 - i_1) + 15 = 0$.

Thus, $(9)i_1 - 3i_2 = 9, \qquad -3i_1 + (7)i_2 = -15.$

Solving, $i_1 = \frac{1}{3} = 0.3333\,\text{A}, \qquad i_2 = -2 = -2\,\text{A}.$

Using passive sign convention, $p_L = -(9)i_1 = 3\,\text{W}$ delivered , $p_R = (15)i_2 = 30\,\text{W}$ delivered .

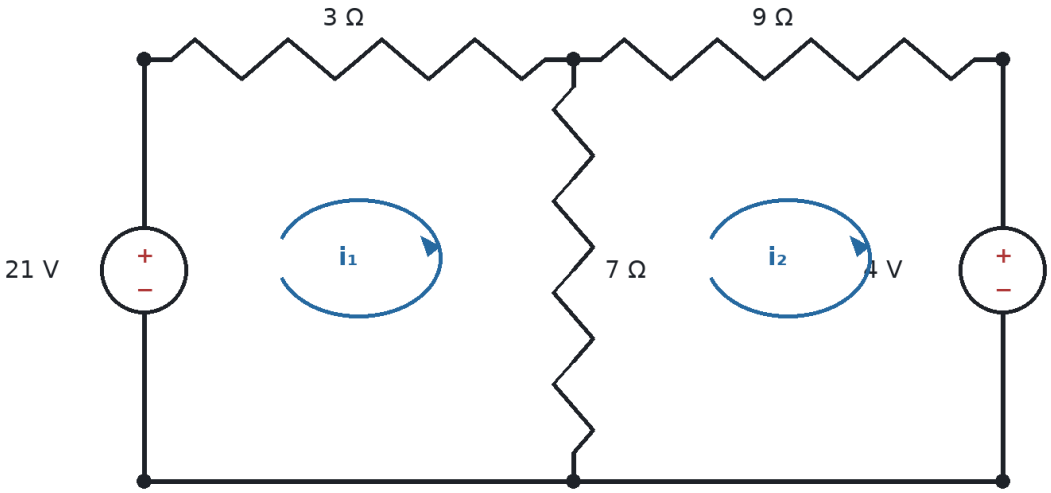
The resistors absorb $P_{\text{res}} = 6i_1^2 + 3(i_1 - i_2)^2 + 4i_2^2 = 33\,\text{W}$.

Therefore, total absorbed power is zero: $p_L + p_R + P_{\text{res}} = 0.$

Question 7

Shared-resistor voltage. Find both mesh currents and the magnitude and direction of current, then calculate the top-to-bottom voltage in the shared resistor. The left top resistor is $3\,\Omega$, the shared resistor is $7\,\Omega$, and the right top resistor is $9\,\Omega$. The left and right sources are $21\,\text{V}$ and $4\,\text{V}$, respectively, both positive at the top.

Shared Voltage



Two adjacent clockwise meshes share a $7\,\Omega$ vertical resistor. The left path has $3\,\Omega$ and a top-positive $21\,V$ source; the right path has $9\,\Omega$ and a top-positive $4\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $3i_1 + 7(i_1 - i_2) - 21 = 0$, $9i_2 + 7(i_2 - i_1) + 4 = 0$.

Thus, $(10)i_1 - 7i_2 = 21, \qquad -7i_1 + (16)i_2 = -4.$

Solving, $i_1 = \frac{308}{111} = 2.775\,\text{A}, \qquad i_2 = \frac{107}{111} = 0.964\,\text{A}.$

The shared current is $i_s = \frac{67}{37} = 1.811\,\text{A}$ downward .

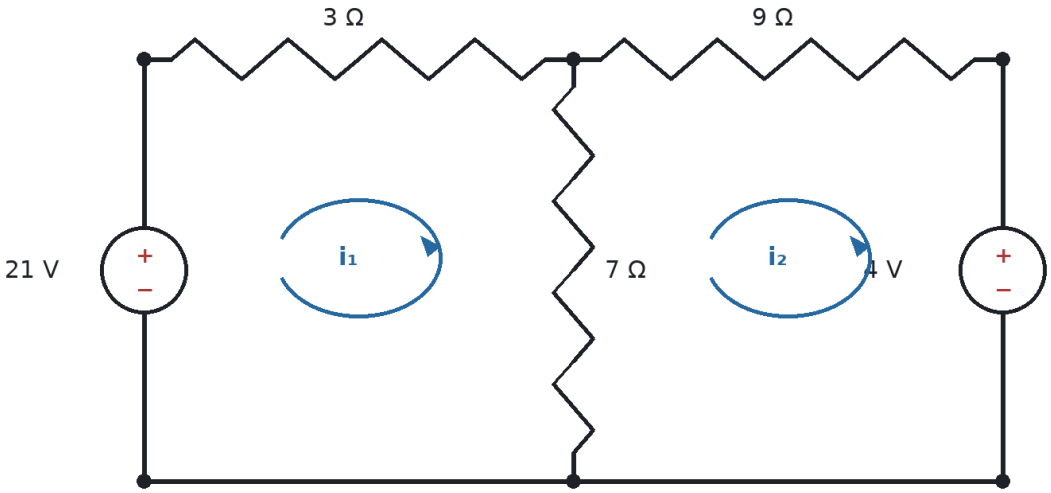
A negative mesh current, if present, flows opposite its clockwise arrow.

The shared-resistor voltage, top relative to bottom, is $v_s = (7\,\Omega)(i_1 - i_2) = 12.68\,\text{V}$.

Question 8

Power in two sources. Find both mesh currents. Determine whether each voltage source absorbs or delivers power, and check total power. The left top resistor is $3\,\Omega$, the shared resistor is $7\,\Omega$, and the right top resistor is $9\,\Omega$. The left and right sources are $21\,\text{V}$ and $4\,\text{V}$, respectively, both positive at the top.

Shared Voltage



Two adjacent clockwise meshes share a $7\,\Omega$ vertical resistor. The left path has $3\,\Omega$ and a top-positive $21\,V$ source; the right path has $9\,\Omega$ and a top-positive $4\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $3i_1 + 7(i_1 - i_2) - 21 = 0$, $9i_2 + 7(i_2 - i_1) + 4 = 0$.

Thus, $(10)i_1 - 7i_2 = 21, \qquad -7i_1 + (16)i_2 = -4.$

Solving, $i_1 = \frac{308}{111} = 2.775\,\text{A}, \qquad i_2 = \frac{107}{111} = 0.964\,\text{A}.$

Using passive sign convention, $p_L = -(21)i_1 = 58.27\,\text{W}$ delivered , $p_R = (4)i_2 = 3.856\,\text{W}$ absorbed .

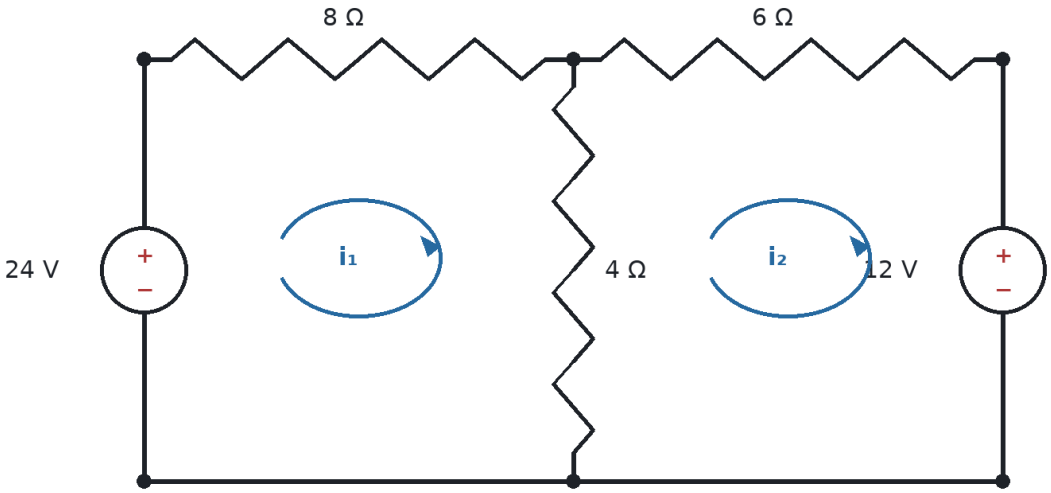
The resistors absorb $P_{\text{res}} = 3i_1^2 + 7(i_1 - i_2)^2 + 9i_2^2 = 54.41\,\text{W}$.

Therefore, total absorbed power is zero: $p_L + p_R + P_{\text{res}} = 0$.

Question 9

Negative right-mesh current. Find both mesh currents and the magnitude and direction of current in the shared resistor. The left top resistor is $8\,\Omega$, the shared resistor is $4\,\Omega$, and the right top resistor is $6\,\Omega$. The left and right sources are $24\,\text{V}$ and $12\,\text{V}$, respectively, both positive at the top.

Source Power



Two adjacent clockwise meshes share a $4\,\Omega$ vertical resistor. The left path has $8\,\Omega$ and a top-positive $24\,V$ source; the right path has $6\,\Omega$ and a top-positive $12\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $8i_1 + 4(i_1 - i_2) - 24 = 0$, $6i_2 + 4(i_2 - i_1) + 12 = 0$.

Thus, $(12)i_1 - 4i_2 = 24, \qquad -4i_1 + (10)i_2 = -12.$

Solving, $i_1 = \frac{24}{13} = 1.846\,\text{A}, \qquad i_2 = \frac{-6}{13} = -0.4615\,\text{A}.$

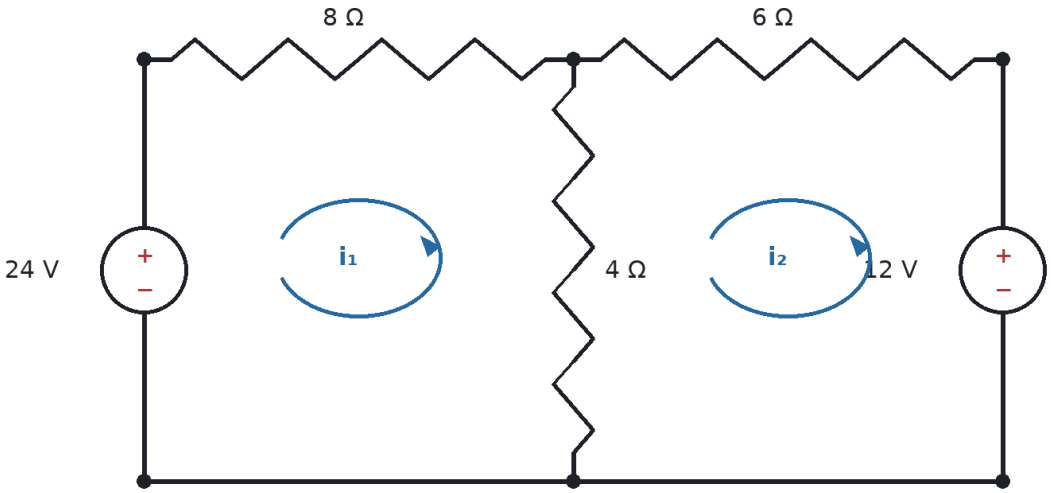
The shared current is $i_s = \frac{30}{13} = 2.308\,\text{A}$ downward .

A negative mesh current, if present, flows opposite its clockwise arrow.

Question 10

Two-mesh resistor power. Find both mesh currents. Determine whether each voltage source absorbs or delivers power, and check total power. The left top resistor is $8\,\Omega$, the shared resistor is $4\,\Omega$, and the right top resistor is $6\,\Omega$. The left and right sources are $24\,\text{V}$ and $12\,\text{V}$, respectively, both positive at the top.

Source Power



Two adjacent clockwise meshes share a $4\,\Omega$ vertical resistor. The left path has $8\,\Omega$ and a top-positive $24\,V$ source; the right path has $6\,\Omega$ and a top-positive $12\,V$ source. One wire joins the bottom nodes.

Solution

Define clockwise mesh currents i_1 and i_2 and sum signed voltage drops. The shared current is downward as $i_s = i_1 - i_2$. $8i_1 + 4(i_1 - i_2) - 24 = 0$, $6i_2 + 4(i_2 - i_1) + 12 = 0$.

Thus, $(12)i_1 - 4i_2 = 24, \qquad -4i_1 + (10)i_2 = -12.$

Solving, $i_1 = \frac{24}{13} = 1.846\,\text{A}, \qquad i_2 = \frac{-6}{13} = -0.4615\,\text{A}.$

Using passive sign convention, $p_L = -(24)i_1 = 44.31\,\text{W}$ delivered , $p_R = (12)i_2 = 5.538\,\text{W}$ delivered .

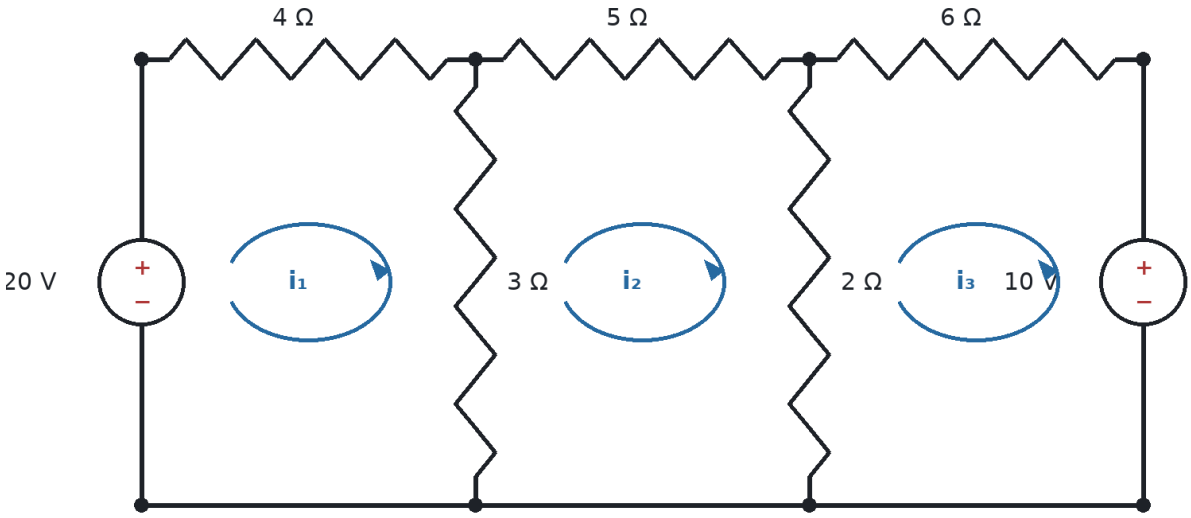
The resistors absorb $P_{\text{res}} = 8i_1^2 + 4(i_1 - i_2)^2 + 6i_2^2 = 49.85\,\text{W}$.

Therefore, total absorbed power is zero: $p_L + p_R + P_{\text{res}} = 0$.

Question 11

Three-mesh equation set. Find all mesh currents and the direction of both shared-branch currents. The top resistors are $4\,\Omega$, $5\,\Omega$, and $6\,\Omega$; shared resistors are $3\,\Omega$ and $2\,\Omega$. The outer sources are $20\,\text{V}$ and $10\,\text{V}$, both positive at the top.

Q14 — Three-Mesh Power Check



Three clockwise meshes have top resistors $4\,\Omega$, $5\,\Omega$, and $6\,\Omega$. Shared branches are $3\,\Omega$ and $2\,\Omega$. Top-positive $20\,V$ and $10\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(7)i_1 - 3i_2 = 20,$$

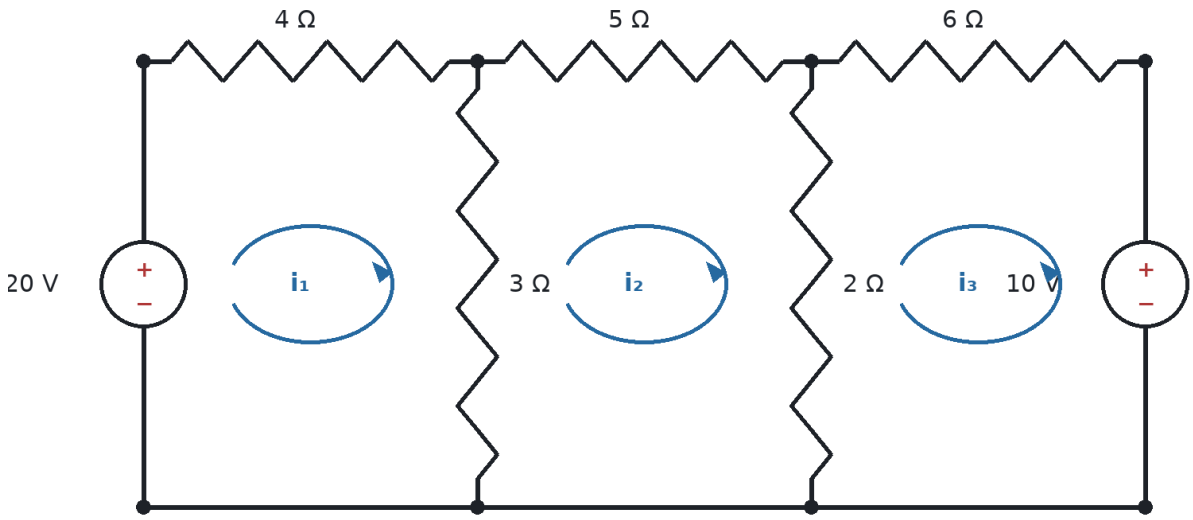
$$-3i_1 + (10)i_2 - 2i_3 = 0,$$

$$-2i_2 + (8)i_3 = -10.$$

Solving, $i_1 = \frac{73}{23} = 3.174\,\text{A}$, $\quad i_2 = \frac{17}{23} = 0.7391\,\text{A}$, $\quad i_3 = \frac{-49}{46} = -1.065\,\text{A}$.
The shared currents are $i_1 - i_2 = \frac{56}{23} = 2.435\,\text{A}$ downward , $i_2 - i_3 = \frac{83}{46} = 1.804\,\text{A}$ downward .
Any negative mesh current flows counterclockwise.

Question 12

Three-mesh power check. Find all mesh currents and verify power balance, including absorbed or delivered source power. The top resistors are $4\,\Omega$, $5\,\Omega$, and $6\,\Omega$; shared resistors are $3\,\Omega$ and $2\,\Omega$. The outer sources are $20\,\text{V}$ and $10\,\text{V}$, both positive at the top.



Three clockwise meshes have top resistors $4\,\Omega$, $5\,\Omega$, and $6\,\Omega$. Shared branches are $3\,\Omega$ and $2\,\Omega$. Top-positive $20\,V$ and $10\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(7)i_1 - 3i_2 = 20,$$

$$-3i_1 + (10)i_2 - 2i_3 = 0,$$

$$-2i_2 + (8)i_3 = -10.$$

Solving, $i_1 = \frac{73}{23} = 3.174\,\text{A}$, $i_2 = \frac{17}{23} = 0.7391\,\text{A}$, $i_3 = \frac{-49}{46} = -1.065\,\text{A}$.

The source powers under passive sign convention are $p_L = -20i_1 = 63.48\,\text{W}$ delivered , $p_R = 10i_3 = 10.65\,\text{W}$ delivered .

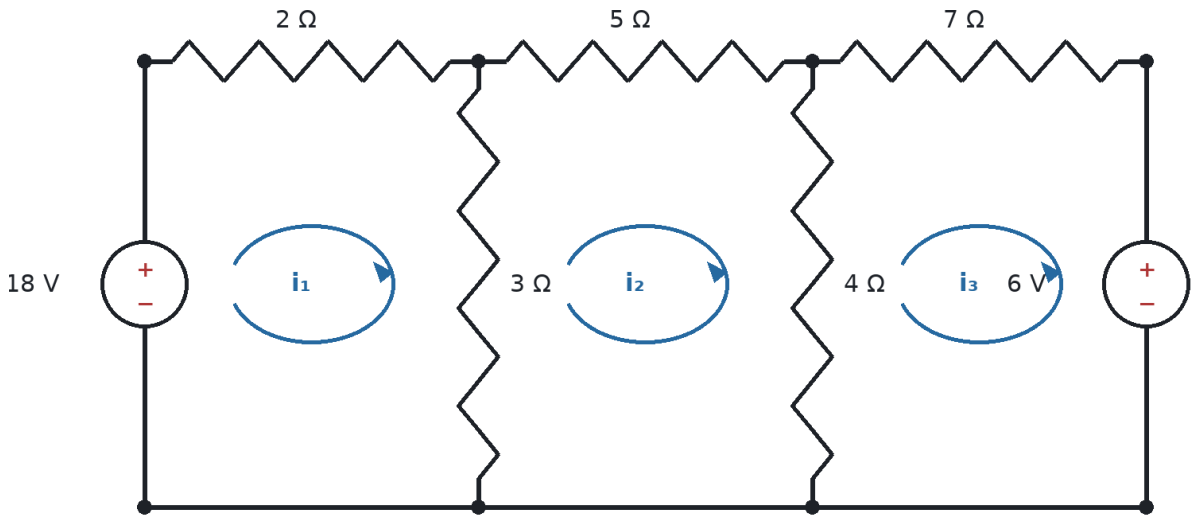
Total resistor absorption is $P_{\text{res}} = 4i_1^2 + 5i_2^2 + 6i_3^2 + 3(i_1 - i_2)^2 + 2(i_2 - i_3)^2 = 74.13\,\text{W}$.

The check is $p_L + p_R + P_{\text{res}} = 0$.

Question 13

Three-mesh coupling. Find all mesh currents and the direction of both shared-branch currents. The top resistors are $2\,\Omega$, $5\,\Omega$, and $7\,\Omega$; shared resistors are $3\,\Omega$ and $4\,\Omega$. The outer sources are $18\,\text{V}$ and $6\,\text{V}$, both positive at the top.

Three Meshes



Three clockwise meshes have top resistors $2\,\Omega$, $5\,\Omega$, and $7\,\Omega$. Shared branches are $3\,\Omega$ and $4\,\Omega$. Top-positive $18\,V$ and $6\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(5)i_1 - 3i_2 = 18,$$

$$-3i_1 + (12)i_2 - 4i_3 = 0,$$

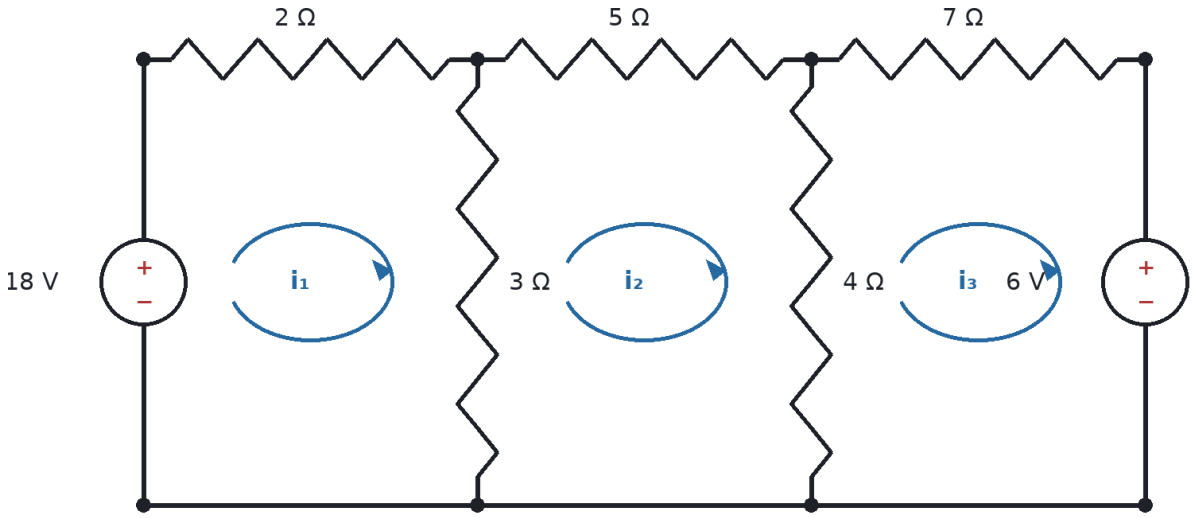
$$-4i_2 + (11)i_3 = -6.$$

Solving, $i_1 = \frac{2016}{481} = 4.191\,\text{A}$, $\quad i_2 = \frac{474}{481} = 0.9854\,\text{A}$, $\quad i_3 = \frac{-90}{481} = -0.1871\,\text{A}$.
The shared currents are $i_1 - i_2 = \frac{1542}{481} = 3.206\,\text{A}$ downward , $i_2 - i_3 = \frac{564}{481} = 1.173\,\text{A}$ downward .
Any negative mesh current flows counterclockwise.

Question 14

Three-mesh source power. Find all mesh currents and verify power balance, including absorbed or delivered source power. The top resistors are $2\,\Omega$, $5\,\Omega$, and $7\,\Omega$; shared resistors are $3\,\Omega$ and $4\,\Omega$. The outer sources are $18\,\text{V}$ and $6\,\text{V}$, both positive at the top.

Three Meshes



Three clockwise meshes have top resistors $2\,\Omega$, $5\,\Omega$, and $7\,\Omega$. Shared branches are $3\,\Omega$ and $4\,\Omega$. Top-positive $18\,V$ and $6\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(5)i_1 - 3i_2 = 18,$$

$$-3i_1 + (12)i_2 - 4i_3 = 0,$$

$$-4i_2 + (11)i_3 = -6.$$

Solving, $i_1 = \frac{2016}{481} = 4.191\,\text{A}$, $i_2 = \frac{474}{481} = 0.9854\,\text{A}$, $i_3 = \frac{-90}{481} = -0.1871\,\text{A}$.

The source powers under passive sign convention are $p_L = -18i_1 = 75.44\,\text{W}$ delivered , $p_R = 6i_3 = 1.123\,\text{W}$ delivered .

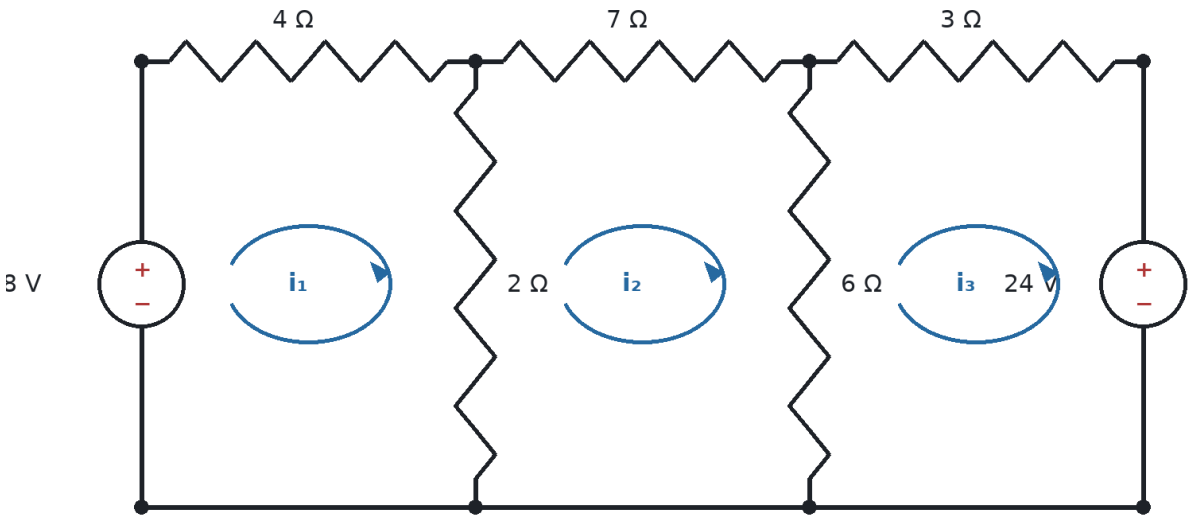
Total resistor absorption is $P_{\text{res}} = 2i_1^2 + 5i_2^2 + 7i_3^2 + 3(i_1 - i_2)^2 + 4(i_2 - i_3)^2 = 76.57\,\text{W}$.

The check is $p_L + p_R + P_{\text{res}} = 0$.

Question 15

Interpret three signed currents. Find all mesh currents and the direction of both shared-branch currents. The top resistors are $4\,\Omega$, $7\,\Omega$, and $3\,\Omega$; shared resistors are $2\,\Omega$ and $6\,\Omega$. The outer sources are $8\,\text{V}$ and $24\,\text{V}$, both positive at the top.

Negative Currents



Three clockwise meshes have top resistors $4\,\Omega$, $7\,\Omega$, and $3\,\Omega$. Shared branches are $2\,\Omega$ and $6\,\Omega$. Top-positive $8\,V$ and $24\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(6)i_1 - 2i_2 = 8,$$

$$-2i_1 + (15)i_2 - 6i_3 = 0,$$

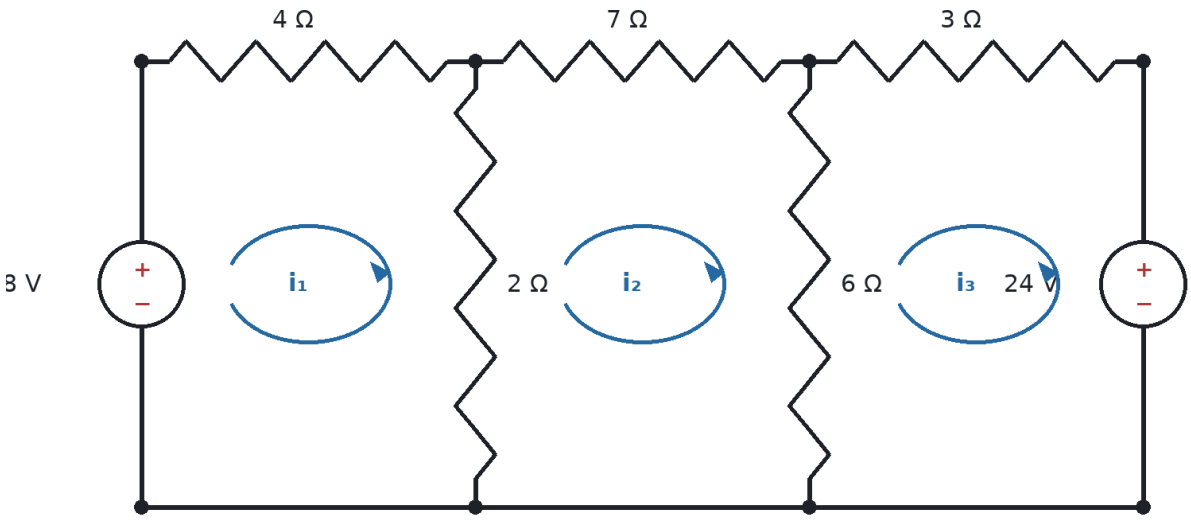
$$-6i_2 + (9)i_3 = -24.$$

Solving, $i_1 = \frac{28}{31} = 0.9032\,\text{A}$, $\quad i_2 = \frac{-40}{31} = -1.29\,\text{A}$, $\quad i_3 = \frac{-328}{93} = -3.527\,\text{A}$.
The shared currents are $i_1 - i_2 = \frac{68}{31} = 2.194\,\text{A}$ downward , $i_2 - i_3 = \frac{208}{93} = 2.237\,\text{A}$ downward .
Any negative mesh current flows counterclockwise.

Question 16

Power with reversed currents. Find all mesh currents and verify power balance, including absorbed or delivered source power. The top resistors are $4\,\Omega$, $7\,\Omega$, and $3\,\Omega$; shared resistors are $2\,\Omega$ and $6\,\Omega$. The outer sources are $8\,\text{V}$ and $24\,\text{V}$, both positive at the top.

Negative Currents



Three clockwise meshes have top resistors $4\,\Omega$, $7\,\Omega$, and $3\,\Omega$. Shared branches are $2\,\Omega$ and $6\,\Omega$. Top-positive $8\,V$ and $24\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(6)i_1 - 2i_2 = 8,$$

$$-2i_1 + (15)i_2 - 6i_3 = 0,$$

$$-6i_2 + (9)i_3 = -24.$$

Solving, $i_1 = \frac{28}{31} = 0.9032\,\text{A}$, $i_2 = \frac{-40}{31} = -1.29\,\text{A}$, $i_3 = \frac{-328}{93} = -3.527\,\text{A}$.

The source powers under passive sign convention are $p_L = -8i_1 = 7.226\,\text{W}$ delivered , $p_R = 24i_3 = 84.65\,\text{W}$ delivered .

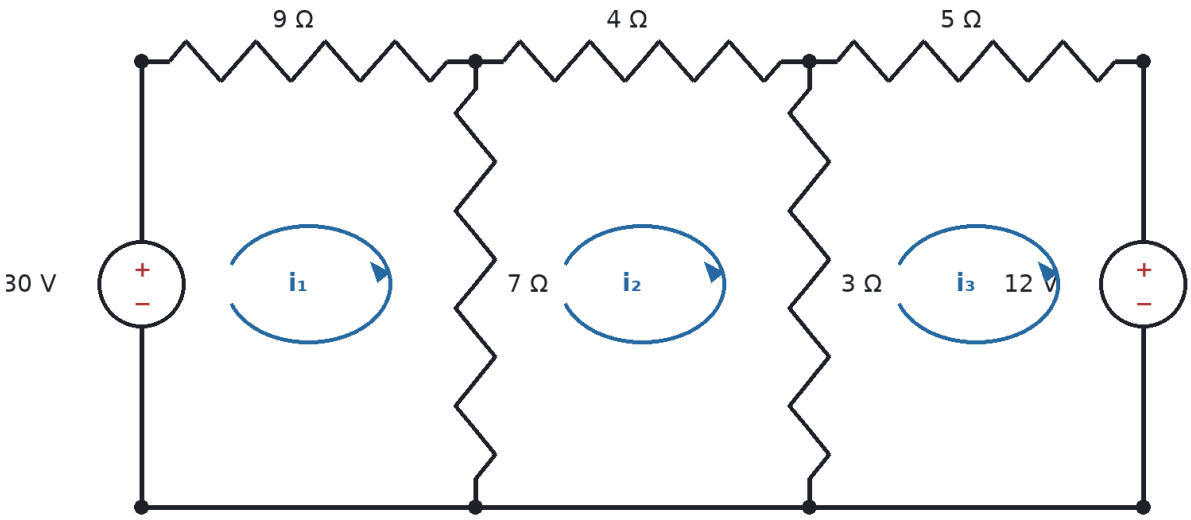
Total resistor absorption is $P_{\text{res}} = 4i_1^2 + 7i_2^2 + 3i_3^2 + 2(i_1 - i_2)^2 + 6(i_2 - i_3)^2 = 91.87\,\text{W}$.

The check is $p_L + p_R + P_{\text{res}} = 0$.

Question 17

Two shared-branch directions. Find all mesh currents and the direction of both shared-branch currents. The top resistors are $9\,\Omega$, $4\,\Omega$, and $5\,\Omega$; shared resistors are $7\,\Omega$ and $3\,\Omega$. The outer sources are $30\,\text{V}$ and $12\,\text{V}$, both positive at the top.

Shared Currents



Three clockwise meshes have top resistors $9\,\Omega$, $4\,\Omega$, and $5\,\Omega$. Shared branches are $7\,\Omega$ and $3\,\Omega$. Top-positive $30\,V$ and $12\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(16)i_1 - 7i_2 = 30,$$

$$-7i_1 + (14)i_2 - 3i_3 = 0,$$

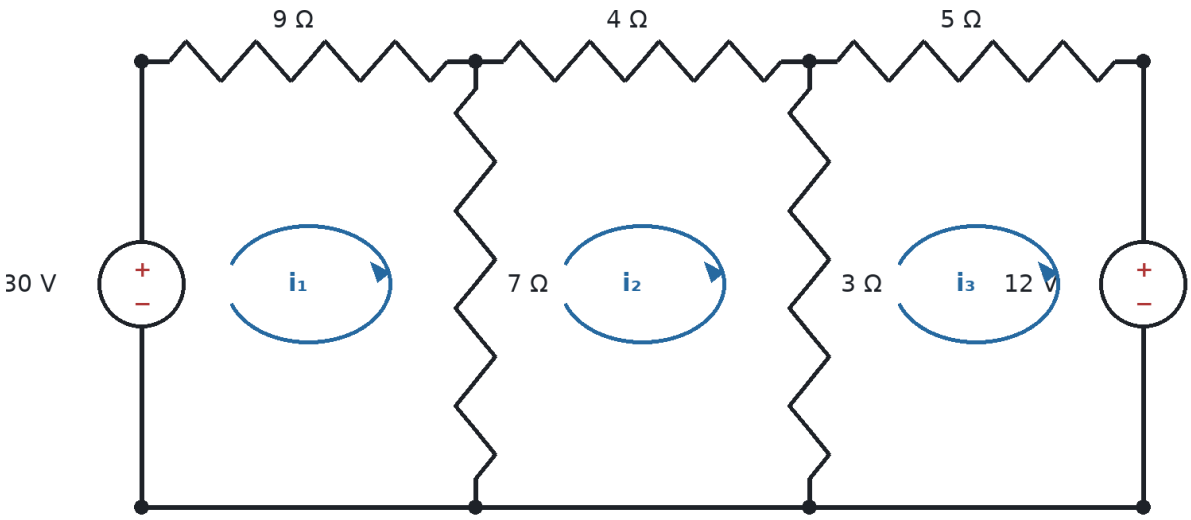
$$-3i_2 + (8)i_3 = -12.$$

Solving, $i_1 = \frac{1419}{628} = 2.26\,\text{A}$, $\quad i_2 = \frac{138}{157} = 0.879\,\text{A}$, $\quad i_3 = \frac{-735}{628} = -1.17\,\text{A}$.
The shared currents are $i_1 - i_2 = \frac{867}{628} = 1.381\,\text{A}$ downward , $i_2 - i_3 = \frac{1287}{628} = 2.049\,\text{A}$ downward .
Any negative mesh current flows counterclockwise.

Question 18

Outer-source power status. Find all mesh currents and verify power balance, including absorbed or delivered source power. The top resistors are $9\,\Omega$, $4\,\Omega$, and $5\,\Omega$; shared resistors are $7\,\Omega$ and $3\,\Omega$. The outer sources are $30\,\text{V}$ and $12\,\text{V}$, both positive at the top.

Shared Currents



Three clockwise meshes have top resistors $9\,\Omega$, $4\,\Omega$, and $5\,\Omega$. Shared branches are $7\,\Omega$ and $3\,\Omega$. Top-positive $30\,V$ and $12\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(16)i_1 - 7i_2 = 30,$$

$$-7i_1 + (14)i_2 - 3i_3 = 0,$$

$$-3i_2 + (8)i_3 = -12.$$

Solving, $i_1 = \frac{1419}{628} = 2.26\,\text{A}$, $i_2 = \frac{138}{157} = 0.879\,\text{A}$, $i_3 = \frac{-735}{628} = -1.17\,\text{A}$.

The source powers under passive sign convention are $p_L = -30i_1 = 67.79\,\text{W}$ delivered , $p_R = 12i_3 = 14.04\,\text{W}$ delivered .

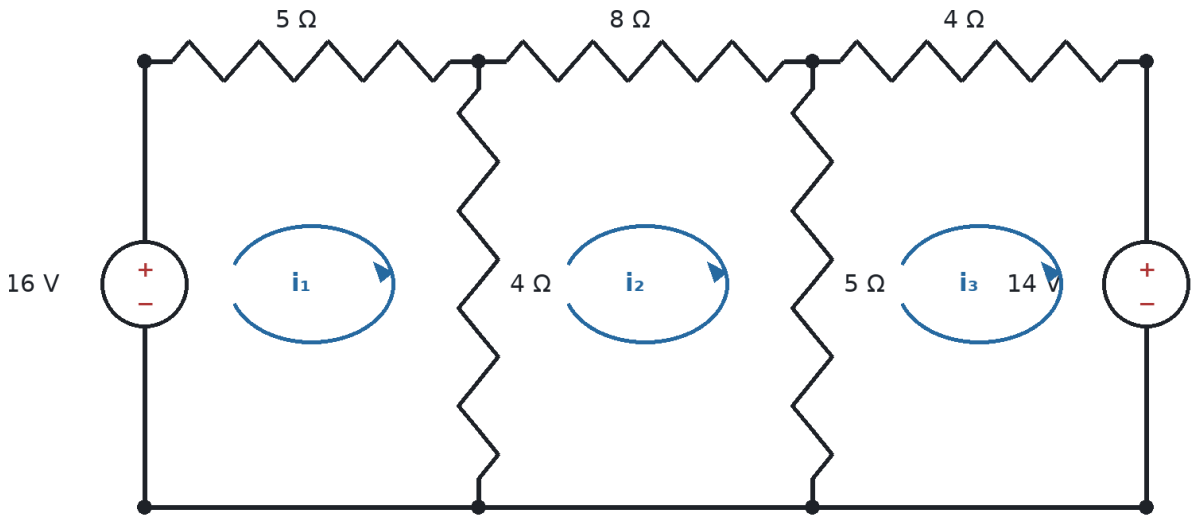
Total resistor absorption is $P_{\text{res}} = 9i_1^2 + 4i_2^2 + 5i_3^2 + 7(i_1 - i_2)^2 + 3(i_2 - i_3)^2 = 81.83\,\text{W}$.

The check is $p_L + p_R + P_{\text{res}} = 0$.

Question 19

Challenge: tightly coupled meshes. Find all mesh currents and the direction of both shared-branch currents. The top resistors are $5\,\Omega$, $8\,\Omega$, and $4\,\Omega$; shared resistors are $4\,\Omega$ and $5\,\Omega$. The outer sources are $16\,\text{V}$ and $14\,\text{V}$, both positive at the top.

Power Balance



Three clockwise meshes have top resistors $5\,\Omega$, $8\,\Omega$, and $4\,\Omega$. Shared branches are $4\,\Omega$ and $5\,\Omega$. Top-positive $16\,V$ and $14\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(9)i_1 - 4i_2 = 16,$$

$$-4i_1 + (17)i_2 - 5i_3 = 0,$$

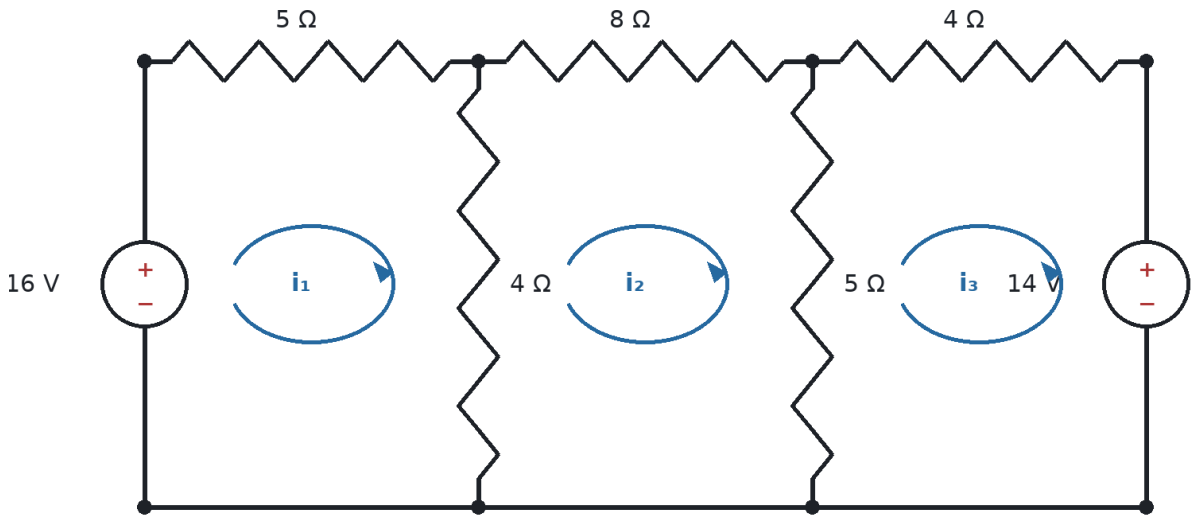
$$-5i_2 + (9)i_3 = -14.$$

Solving, $i_1 = \frac{221}{126} = 1.754\,\text{A}$, $i_2 = \frac{-3}{56} = -0.05357\,\text{A}$, $i_3 = \frac{-799}{504} = -1.585\,\text{A}$.
The shared currents are $i_1 - i_2 = \frac{911}{504} = 1.808\,\text{A}$ downward , $i_2 - i_3 = \frac{193}{126} = 1.532\,\text{A}$ downward .
Any negative mesh current flows counterclockwise.

Question 20

Challenge: full power balance. Find all mesh currents and verify power balance, including absorbed or delivered source power. The top resistors are $5\,\Omega$, $8\,\Omega$, and $4\,\Omega$; shared resistors are $4\,\Omega$ and $5\,\Omega$. The outer sources are $16\,\text{V}$ and $14\,\text{V}$, both positive at the top.

Power Balance



Three clockwise meshes have top resistors $5\,\Omega$, $8\,\Omega$, and $4\,\Omega$. Shared branches are $4\,\Omega$ and $5\,\Omega$. Top-positive $16\,V$ and $14\,V$ sources close the left and right sides; the bottom is one wire.

Solution

Define clockwise mesh currents i_1, i_2, and i_3. Downward shared currents are $i_1 - i_2$ and $i_2 - i_3$.

$$(9)i_1 - 4i_2 = 16,$$

$$-4i_1 + (17)i_2 - 5i_3 = 0,$$

$$-5i_2 + (9)i_3 = -14.$$

Solving, $i_1 = \frac{221}{126} = 1.754\,\text{A}$, $i_2 = \frac{-3}{56} = -0.05357\,\text{A}$, $i_3 = \frac{-799}{504} = -1.585\,\text{A}$.

The source powers under passive sign convention are $p_L = -16i_1 = 28.06\,\text{W}$ delivered , $p_R = 14i_3 = 22.19\,\text{W}$ delivered .

Total resistor absorption is $P_{\text{res}} = 5i_1^2 + 8i_2^2 + 4i_3^2 + 4(i_1 - i_2)^2 + 5(i_2 - i_3)^2 = 50.26\,\text{W}$.

The check is $p_L + p_R + P_{\text{res}} = 0$.